Dmitry Glukhov

AI / Software Engineer | Mathematical Reasoning Systems

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PhD

Profile

AI / Software Engineer building evidence-driven AI, mathematical reasoning and research products. I turn complex source material, model outputs and noisy datasets into inspectable systems with explicit provenance, durable state, evaluation and useful interfaces. My recent work spans Wolfram-verified problem solving, temporal evidence systems, LLM evaluation, historical media reconstruction and scientific pipelines.

Selected Projects

Wolfram Agent - AI-assisted mathematical reasoning | Local R&D

- Built a local mathematics assistant that turns text and image problems into worked solutions with mandatory Wolfram verification, PDF reports and plots.
- Applied Reduce, FullSimplify, Resolve / ForAll and FindInstance to equivalence checks, quantified constraints, integer feasibility and counterexample search.
- Produced 46 worked mathematics reports and a safe execution layer with hard timeouts; computational checks and human-readable reasoning remain visibly distinct.

Reality Wiki - temporal evidence and model memory | Private R&D

- Built an append-only epistemic ledger that separates publication, availability, ingestion and event time, with historical and operational as-of replay and SHA-256 evidence hashes.
- Mirrored and normalized official CBR and Rosstat sources into cutoff-safe SQLite, DuckDB and Parquet layers with explicit provenance and fail-closed source boundaries.
- Implemented a 26-origin walk-forward memory study with 104 resolved questions per arm and a stricter forward-only experiment. Retrospective results are explicitly not presented as forecasting skill.

Time Receiver - historical media PWA | Live product | Public release

- Built a static-first radio and television environment for Moscow across 18,628 days from 1940 to 1990.
- Packaged 4.09M radio and 395K TV events into 51 on-demand yearly bundles with an offline-capable PWA shell.
- Designed source-aware fidelity labels for exact schedules, reconstructed gaps and equivalent media, plus archival ingestion and editorial-review pipelines.

Form / Field - executable generative graphics laboratory | Live lab | Source

- Built a Vue/Vite/p5.js research interface around 34 clearly attributed sketches and a code snapshot of 845 community works by 26 authors.
- Made the visible RAW code the actual rendering source; added bounded genome compilation, manual 3D projection, topology tools and stateful entities.
- Shipped responsive lab, archive, evolution and theory surfaces with 16 Node test modules and automated Pages deployment.

Native Audio Translator | Repository

- Built a native macOS prototype for live audio translation and OCR subtitle overlays with separate capture, recognition, stabilization, translation and rendering stages.
- Added runtime permission flows, command control, validation executables, local diagnostics and app bundling.

Additional Public Work

- Feynman Reader - bilingual static reader for 3 volumes and 115 chapters, with 12,644 formulas, 1,038 figures, media manifests and validation tooling.

Skills

Wolfram Language, Python, TypeScript, Swift, Rust, Vue, React, SQLite, DuckDB, Parquet, Docker, GitHub Actions, LLM pipelines, symbolic verification, quantified constraints, counterexample search, temporal replay, model evaluation, OCR, speech-to-text and scientific computing.